

Image Processing

Total points 19/19

The respondent's email (**aaradhyadevtmr@gmail.com**) was recorded on submission of this form.

✓ OpenCV loads images in BGR. You display one directly with `plt.imshow()` *1/1 without converting. What do you see?

- ☐ Rotated 90 degrees
- ☒ R and B channels swapped
- ☐ Greyscale
- ☐ Vertically flipped



✓ You want to isolate a red apple from its background regardless of whether the warehouse lighting is dim or bright. Which color space is most useful, and why? *1/1

- ☐ RGB
- ☐ Grayscale
- ☐ BGR
- ☒ HSV



✓ Red has two HSV ranges in OpenCV: roughly [0,10] and [170,180]. Why does red need two ranges when green or yellow needs only one? *1/1

- ☐ Red objects have higher saturation than other colors, requiring two separate saturation bands
- ☒ OpenCV's HSV range is [0,180], and red's hue wraps around the circular axis at both ends, so a single contiguous range would exclude some true reds ✓
- ☐ The two ranges handle underripe vs. overripe red apples separately
- ☐ cv2.inRange() can only accept ranges up to 90 wide, so red must be split for implementation reasons

✓ A histogram of a grayscale fruit image shows almost all pixels clustered near value 30–50 out of 255. What does this tell you? *1/1

- ☐ High Contrast
- ☒ Overexposed ✓
- ☐ Underexposed
- ☐ The image was converted from HSV, which clips values above 50



✓ CLAHE is applied with clipLimit=2.0 and again with clipLimit=8.0. At clipLimit=8.0 the fruit edges become noisier. What does this indicate about the trade-off? *1/1

- ☐ Higher clipLimit means less equalization, so 8.0 should always produce less noise than 2.0
- ☒ clipLimit caps how much any histogram bin can be redistributed; a high cap aggressively boosts low-contrast regions but also amplifies noise that was previously invisible ✓
- ☐ CLAHE's tileGridSize parameter, not clipLimit, controls noise amplification
- ☐ The noisier output at 8.0 means the image was already well-exposed and CLAHE shouldn't have been applied

✓ You implement conv2d() from scratch and get a slightly different result from cv2.filter2D(). What is the most likely explanation? *1/1

- ☐ cv2.filter2D() uses FFT-based convolution internally, which introduces rounding error
- ☐ Your implementation uses a different kernel size
- ☒ Convolution flips the kernel 180° before sliding; cross-correlation doesn't flip. Results differ for asymmetric kernels like Sobel. ✓
- ☐ float32 and uint8 give identical results in both, so any difference must be a bug



✓ You add Gaussian noise to an orange image and compare Gaussian blur, *1/1
median filter, and bilateral filter. For Gaussian noise, which best preserves
fruit edges?

- ☐ Gaussian Blur
- ☐ Median Filter
- ☒ Bilateral Filter
- ☐ All three perform identically on Gaussian noise by definition
- ☐ Option 5



✓ Erosion is applied to a binary banana HSV mask with a 5×5 kernel. What *1/1
happens to small noise blobs and to the banana's main region?

- ☐ Small blobs grow; main region shrinks
- ☒ Both small blobs and the main region shrink
- ☐ Small blobs disappear; main region is unaffected
- ☐ Only bright pixels are affected; dark background pixels stay the same



✓ What is the practical difference between Opening and Closing? * 1/1

- ☐ Opening fills holes; Closing removes noise
- ☒ Opening removes small white noise blobs; Closing fills small holes in white foreground regions
- ☐ They are the same operation applied in different order with no practical difference
- ☐ Opening works on grayscale images; Closing only works on binary masks



✓ You apply the morphological gradient (Dilation – Erosion) to a binary fruit mask. What does the result represent? *1/1

- ☐ The filled interior of the fruit
- ☒ The outline (boundary) of the fruit object ✓
- ☐ The background with the fruit removed
- ☐ A noise-free version of the original mask

✓ In the Canny pipeline, what is the purpose of Non-Maximum Suppression (NMS)? *1/1

- ☐ It removes pixels below the low threshold
- ☐ It links weak edges to strong edges using hysteresis
- ☒ It thins wide gradient ridges to single-pixel-wide edges by keeping only local maxima along the gradient direction ✓
- ☐ It applies Gaussian smoothing before gradient computation

✓ You run `cv2.Canny()` with thresholds `[30, 100]` and `[100, 200]`. The `[30, 100]` result has more edge pixels inside the fruit. Why? *1/1

- ☐ A lower high threshold makes fewer pixels qualify as strong edges, so fewer weak edges survive hysteresis
- ☒ A lower low threshold admits more weak edge pixels ✓
- ☐ Thresholds have no effect on internal texture
- ☐ The `[30, 100]` run applied a larger Gaussian blur kernel, which smoothed the fruit interior



✓ Harris Corner Detection uses eigenvalues of the gradient covariance matrix M at each pixel. What do two large eigenvalues indicate? *1/1

- ☐ A flat region with no gradient in any direction
- ☐ An edge
- ☒ A corner
- ☐ A noise artifact from Gaussian pre-smoothing



✓ You lower the Harris response threshold from 0.1 to 0.01. What changes? *1/1

- ☐ Fewer corners detected
- ☒ More pixels qualify as corners
- ☐ Corner positions shift but count stays the same
- ☐ Only the display color of the corner markers changes



✓ In the Hough Circle Transform on mixed_bowl.jpg, you increase param2 (the accumulator threshold). What is the effect? *1/1

- ☐ More circles detected
- ☐ Minimum and maximum detectable radius both increase
- ☒ Fewer circles detected
- ☐ param2 controls Gaussian blur strength, not detection sensitivity



✓ A student applies morphological operations directly to a BGR fruit image instead of the binary mask. What most likely happens? *1/1

- ☐ Identical results to applying it on the binary mask
- ☒ Each color channel is eroded/dilated independently; the result is a color-distorted image rather than a cleaned binary region ✓
- ☐ A runtime error, since morphological operations only accept binary input
- ☐ The image is converted to grayscale automatically before the operation runs

✓ Sobel X highlights vertical edges and Sobel Y highlights horizontal edges. 1/1
What does computing $\sqrt{S_x^2 + S_y^2}$ give you?

- ☐ A directional edge map showing only horizontal contours
- ☒ An edge strength map sensitive to edges at any orientation ✓
- ☐ The same result as applying a box blur
- ☐ A binary mask separating fruit from background

✓ After running the full fruit pipeline, the bounding box encloses a large noisy region rather than just the fruit. Which step most likely failed? *1/1

- ☐ Canny thresholds are too low
- ☐ HSV range is too narrow
- ☒ Morphological cleanup is insufficient ✓
- ☐ `safe_imwrite()` is saving the wrong image; the pipeline result is correct



- ✓ What is the main takeaway connecting today's manual filters to the CNN module's `nn.Conv2d` layers? *1/1
- ☐ CNNs and classical CV use completely different mathematical operations and cannot be compared
 - ☒ The `Conv2d` operation is identical to the manual `conv2d()` — the only difference is that CNN kernels are learned from data while today's kernels (Sobel, Gaussian, sharpen) are hand-designed ✓
 - ☐ Classical filters are always better than learned CNN filters because they have no training overhead
 - ☐ `nn.Conv2d` applies Sobel X and Sobel Y by default; today's session showed what those look like

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